

Voxa AI: Project Presentation Guide

A Complete Guide to Features, Architecture, Demonstration Script, and Q&A;

1. Project Overview: What is Voxa AI?

Voxa AI is a professional, enterprise-grade AI translation suite that integrates real-time audio capture, document translation, and speech synthesis into a unified workspace.

Unlike generic translation utilities, Voxa AI is built for corporate presentation readiness, providing real-time streaming translation of spoken language, a multi-stage PDF reading assistant, and default language profiles to personalize user workspace sessions.

2. Key Product Features

A. Real-Time Speech Translation Workspace

- **Acoustic Capture:** Captures microphone input directly from the browser using the Web Audio API.
- **Streaming Updates:** Employs Server-Sent Events (SSE) to update the client checklist (Uploading → Transcribing → Translating → Synthesizing → Finished) step-by-step in real time.
- **Text-to-Speech (TTS):** Translates input and generates corresponding natural speech audio using ElevenLabs TTS.

B. Dedicated AI PDF Reading Assistant

- **Text Extraction:** Extracts textual contents from uploaded PDF documents in-memory.
- **Smart Text Chunking:** Automatically tokenizes translated text into sentence-aware blocks of 3,000–5,000 characters to stay within ElevenLabs' API limits, preventing character quota crashes.
- **Live Progress Stream:** Displays processing updates (e.g. 'Reading page 1...', 'Generating audio part 2/5...') using line-delimited streaming buffers.
- **TTS Merger & Fallback:** Sequentially builds and merges MP3 files. If ElevenLabs limits are hit, it outputs translations with a clear warning alert instead of crashing.

C. Secure Multi-Route Authentication & Profiles

- **Route Protection:** Protects routes like /workspace, /profile, and /pdf-reader, redirecting unauthenticated users immediately to /login.
- **Language Preferences:** Persists and syncs user's default source and target languages across pages.

3. Technical Stack & Architecture

Frontend: Built with Next.js, React, TypeScript, and Tailwind CSS (v4). Manages raw MediaRecorder audio streams and parses NDJSON response buffers.

Backend: Powered by FastAPI (Uvicorn), providing asynchronous controllers and SSE stream handlers.

Data Store: SQLite DB mapped via SQLAlchemy ORM for storing credentials and default profile settings.

APIs Integrated: ElevenLabs TTS, PyPDF (text parsing), and Resend API (Sandbox mailer OTP fallback).



4. Live Demo Presentation Script (5 Minutes)

Minute 0–1: Introduction & Problem Statement

"Good morning trainers. Today, I am presenting Voxa AI. The core problem we are solving is the language barrier in corporate meetings and document workflows. Voxa AI provides an enterprise SaaS solution for voice and document translation."

Minute 1–2: Landing Page & Auth Flow

[Show Landing Page, click Workspace, show login redirect, log in, show profile page]

"Our workspace pages are protected. Attempting to access them redirect us to login. Once logged in, our workspace automatically adapts itself to my default language preferences: English to Hindi."

Minute 2–3: Speech Workspace Demo

[Click mic, record a phrase, click stop, show checklist progress updates, click Play TTS]

"Notice the checklist updates in real-time. Once translated, I can play the synthesized speech. Clicking Clear Cache stops active playbacks instantly and resets the pipeline."

Minute 3–4: PDF Assistant Demo

[Go to /pdf-reader, upload PDF, translate, show progress streaming texts]

"For large PDFs, we prevent ElevenLabs character crashes by chunking text into 3k-5k character blocks. The button streams real-time steps. If ElevenLabs limits are hit, we output translation text with a warning alert instead of crashing."

5. Trainer Q&A; (Cheat Sheet)

Q1: How did you solve the ElevenLabs 10k character limit?

A: "We implemented a paragraph/sentence boundary-aware text splitter. It breaks text into 3,000–5,000 character blocks, never cutting words in half. Chunks are synthesized sequentially and merged into output.mp3 before returning the URL."

Q2: What happens if ElevenLabs character quota runs out?

A: "We wrapped chunk synthesis in a try-except block. If a quota exception is caught, the backend returns the translations along with a warning payload. The frontend displays the translations immediately and alerts the user with a red warning banner: 'ElevenLabs API Limit Over', maintaining a good user experience."

Q3: How is real-time progress shown on the PDF page?

A: "We use FastAPI's StreamingResponse to yield JSON lines. The frontend reads these chunks incrementally using a ReadableStream reader, updating the status text inside the button layout dynamically."

Q4: Why is the sidebar separate from /workspace?

A: "To keep the original Speech Translation workspace clean and focused, we separated the routes. The workspace page is purely dedicated to speech streams, and the PDF Reading Assistant is hosted at /pdf-reader."